

DEEPSEEK-AI

DeepSeek-V2: Efficient MoE Language Model with Multi-Head Latent Attention

236B total / 21B activated • 128K context • A Strong, Economical Architecture

The LLM Scaling Dilemma:

Performance vs. Training Cost and Inference Speed

Scaling laws demand larger models for better intelligence, but this introduces critical bottlenecks.

The Intelligence Gain

Larger parameter counts yield better reasoning and knowledge, but incur prohibitive compute and memory training costs.

The Inference Bottleneck

The KV cache grows linearly with context length and batch size, throttling generation speed and max concurrent requests.

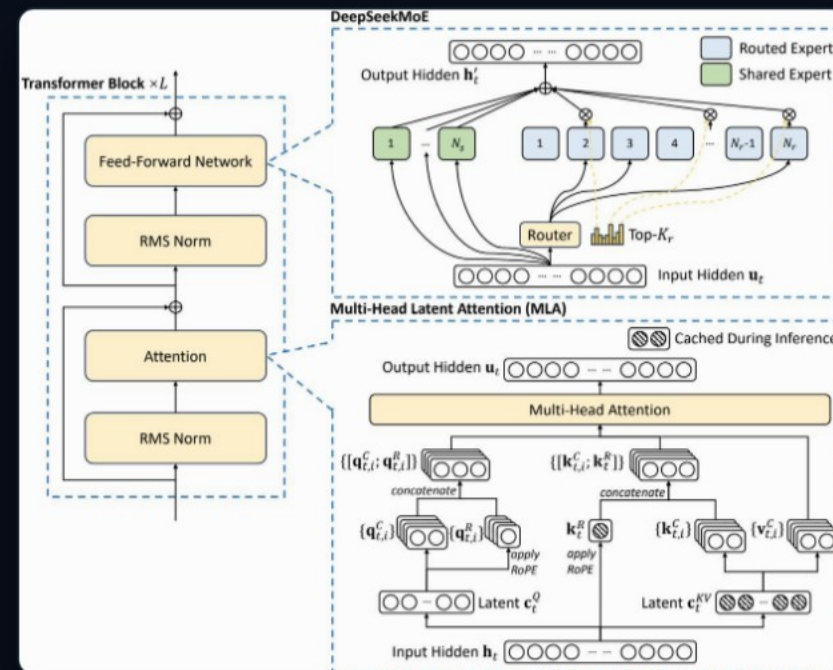
The Requirement

We need high capability achieved through sparse activation (MoE) and fundamentally efficient attention mechanisms.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE

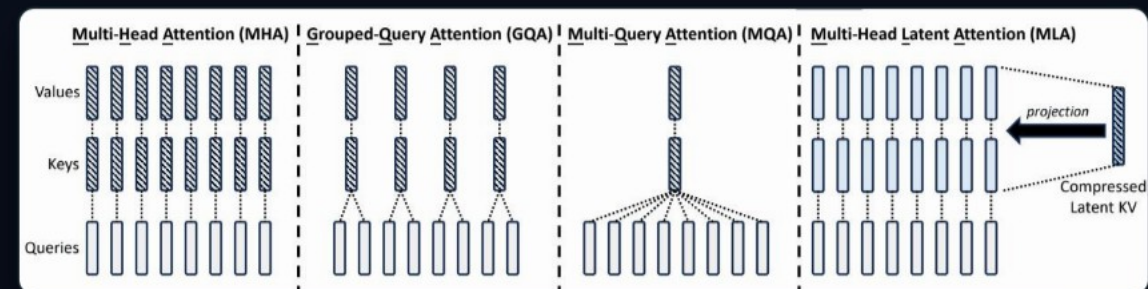
A transformed block design built for radical efficiency during both training and inference.

- Replaces standard Multi-Head Attention with **Multi-Head Latent Attention (MLA)** to slash memory overhead.
- Replaces dense FFN with **DeepSeekMoE** within each Transformer block.
- **Result:** Highly efficient forward pass, requiring minimal activated parameters per token.



Multi-Head Latent Attention Compresses KV Cache by 93.3%

- **The Problem:** MHA stores full Key-Value pairs. GQA/MQA reduce cache but sacrifice model performance.
- **The Solution:** Low-rank key-value joint compression. K and V are compressed into a tiny latent vector.
- **Decoupled RoPE:** Positional keys are handled separately to maintain precise attention localization.
- **The Result:** Better performance than MHA with radically smaller cache requirements during autoregressive generation.



DeepSeekMoE: Fine-Grained Experts with Shared Isolation

Balancing general knowledge retention with highly specialized routing efficiency.



Shared Experts

A set of experts that are **always activated** for every token to capture common, generalized knowledge.



Routed Experts

Sparsely activated via a **top-K router**. Fine-grained segmentation allows for deeper domain specialization.



Device-Limited Routing

Constrains token routing to at most M devices, severely **reducing cross-node communication** overhead.

Efficiency vs. DeepSeek 67B

Massive cost reductions through architectural innovation

42.5%

TRAINING COST
REDUCTION

93.3%

KV CACHE SIZE
REDUCTION

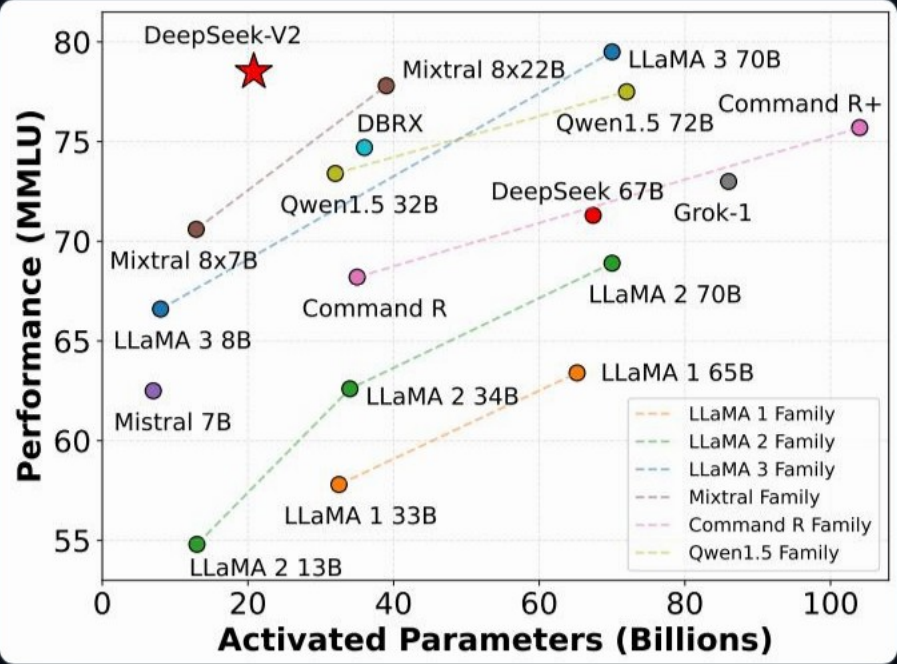
5.76×

MAX GENERATION
THROUGHPUT

Top MMLU Among Open-Source Models

DeepSeek-V2 achieves Pareto-optimal efficiency: matching the capability of 70B-class models using only 21B activated parameters.

MODEL	ACTIVATED PARAMS (B)	MMLU SCORE
DeepSeek-V2	21	78.5
Mixtral 8x22B	39	77.6
LLaMA 3 70B	70	77.0
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0



DeepSeek-V2 Chat (RL): Rivals Closed-Source Models

Alignment Pipeline: SFT on 1.5M conversational sessions followed by GRPO Reinforcement Learning.

38.9%

ALPACAEVAL 2.0
(Length-controlled win rate)

8.97

MT-BENCH
(Overall score)

7.91

ALIGNBENCH
(Outperforms all open-source in Chinese)

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

- **MLA is a Paradigm Shift:** A strictly better alternative to standard Multi-Head Attention, reducing KV cache by 93.3% without sacrificing performance.
- **Economical Training via DeepSeekMoE:** Fine-grained experts and shared expert isolation enable training a 236B model at 42.5% lower cost than a 67B dense model.
- **Unmatched Parameter Efficiency:** Only 21B activated parameters needed to match or exceed 70B-class dense models on comprehensive benchmarks like MMLU.
- **Deployment Ready:** The massive 5.76× throughput improvement makes serving a top-tier open-source model highly practical at scale.
- **World-Class Alignment:** DeepSeek-V2 Chat (RL) leads open-source models in Chinese and remains highly competitive globally in English open-ended evaluation.